



Amrita School of AI
COIMBATORE CAMPUS

25AID432 · MATHEMATICAL POSTULATES OF QUANTUM MECHANICS

Course Handbook

How this course runs, what you will be able to do at the end of it, and how the marks are earned.

Prepared by **Dr. Abhijith Anandakrishnan**

Assistant Professor, Amrita School of AI

COURSE

25AID432 Mathematical Postulates of
Quantum Mechanics

STRUCTURE

Micro-credential, 1 credit

CONTACT

One lecture hour a week

DURATION

15 teaching weeks

OFFERING

Odd Semester 2026-27

What this course is about

Quantum mechanics has a reputation for being strange. Most of that reputation comes from the physics. The mathematics underneath it is small, precise and learnable in one semester: complex numbers, vectors and matrices, and four rules that say how those objects describe a physical system.

This course teaches exactly that mathematics, and nothing you do not need. By the end of it you will be able to take a one- or two-qubit system, write its state as a vector, apply operations to it as matrices, and predict the probability of every measurement outcome by hand. That is the working language of the rest of your degree: quantum computing, quantum communication and quantum sensing all assume it on day one.

The course has three units, in the order the syllabus gives them.

UNIT	TITLE	SESSIONS	WHAT IT GIVES YOU
1	Complex numbers	1 to 3	The numbers quantum amplitudes are made of, and the idea of phase
2	Linear algebra	4 to 9	Vectors, matrices, inner products and the special matrices quantum mechanics uses
3	The postulates of quantum mechanics	10 to 15	Bra - ket notation, the qubit, measurement, operations, many qubits, observables

Units 1 and 2 are tools. Unit 3 is where they are used, and it will feel much easier than it looks if the first two are solid.

NOTE · TWO TOPICS ARE ADDED TO THE PRINTED SYLLABUS

Eigenvalues and eigenvectors, and the **tensor product**, are taught at the end of Unit 2 even though the syllabus does not list them there. Unit 3 cannot be done without them: measurement outcomes of an observable are eigenvalues, and two qubits together are described by a tensor product. They are marked "Beyond the syllabus" in the notes, and they are examinable.

Learning outcomes

OUTCOME	
CO1	Perform arithmetic on complex numbers in Cartesian and polar form, and use Euler's formula to interpret modulus and phase
CO2	Compute with complex vectors and matrices, including inner products, norms, orthonormal bases, projections, and Hermitian and unitary matrices
CO3	State the postulates of quantum mechanics and express states, measurements and operations in bra - ket notation
CO4	Calculate measurement probabilities, post - measurement states, expectation values and the action of unitary operations on one - and two - qubit systems

Assessment

This is a **micro-credential**, and it carries three components.

COMPONENT	WEIGHT	INTERNAL OR EXTERNAL
Class assignment, "The Qubit Workbook"	50%	Internal
End-semester examination	30%	External
Class examination	20%	Internal

NOTE · THIS SUPERSEDES THE GENERIC LINE IN THE CURRICULUM DOCUMENT

The 2025 curriculum booklet states only "internal 70, external 30". The three components above are how that split is made up for this course.

Class assignment, 50%

One workbook, built up across the semester in three parts, one per unit, and closed by a short viva.

PART	COVERS	CLASSROOM 50 ITEM	MARKS
A	Complex numbers: a handwritten problem set, with NumPy checks of your answers	<code>mpqt-a</code>	10
B	Linear algebra: a handwritten problem set, with NumPy checks	<code>mpqt-b</code>	14
C	A small qubit simulator in NumPy (auto-checked), and a short report of three experiments run with it	<code>mpqt-c</code>	20
Viva	Five minutes: explain any line of your code and any step of Parts A and B	In class, after Part C closes	6
Total			50

Every part is submitted through **Classroom 50**, classroom `qts-mpqt` : each part is a private repository created for you when you accept it. Deadlines are set on Classroom 50 and announced in class. The Getting Started guide shows every step, and starts with a practice item, A0, that carries no marks.

Parts A and B are done by hand because the skill being assessed is doing the calculation. The NumPy check that goes with each is there so you find your own mistakes before submitting.

Part C asks you to write, in about a hundred lines of Python, the thing Unit 3 describes on paper: a state as an array, gates as matrices, the Born rule as a function, and measurement as a random draw. When your own code reproduces the predictions you worked out by hand, the postulates stop being abstract.

PITFALL · DO NOT LEAVE PART C TO THE LAST FORTNIGHT

Every function in Part C corresponds to one session of Unit 3. Students who write each function the week it is taught reach the deadline with nothing left to do but the report. Students who start in the final week are learning measurement, operations, tensor products and observables at once.

Class examination, 20%

One multiple-choice test on the Evolveus platform, held in a scheduled test slot after week 9. It covers **Units 1 and 2**, including eigenvalues and tensor products. Most questions require a calculation; a few test whether you can spot a common mistake. One or more options may be correct, and a wrong option costs a quarter of the question's marks.

End-semester examination, 30%

A written paper on the whole syllabus. Unit 3 carries the most weight. Expect one longer problem that takes a small system through all four postulates: prepare a state, apply operations, measure, and interpret the result.

Week-by-week plan

Weeks are numbered from the start of instruction, so this plan is reusable in any offering. There is one session per week.

WEEK	SESSION TOPIC	DECK	ASSIGNMENT
1	Why quantum mechanics needs complex numbers; what a complex number is; the complex plane	q01	Part A released
2	Arithmetic: add, multiply, conjugate, modulus, divide	q01	
3	Euler's formula, polar form, multiplication as rotation, phase	q02	
4	Vectors in $\mathbb{C}^n$; transpose and conjugate transpose	q03	Part A can be completed
5	Matrices: products, non-commutativity, identity, inverse	q03	
6	Inner product, norm, normalisation, orthogonality	q04	
7	Bases, orthonormal bases, inner product as projection	q04	
8	Special matrices: Hermitian and unitary	q05	
9	Eigenvalues and eigenvectors; the tensor product	q05	
10	The postulates; bra-ket notation	q06	Part B can be completed. Class exam in a test slot around this week
11	The qubit; superposition; global and relative phase	q06	Part C: states and inner products
12	Quantum measurement: the Born rule and collapse	q07	Part C: probabilities and measurement
13	Quantum operations: unitary evolution and the standard gates	q07	Part C: gates
14	Multiple quantum states: two qubits, CNOT, entanglement	q08	Part C: tensor products
15	Observables and expectation values; course review	q08	Part C can be completed; viva after it closes

Working in this course

How to use the material

For every session there is a **slide deck** and a section of a **notes booklet**. The decks are what is shown in class. The notes are complete: every definition, every worked example, and a set of self-assessment questions with full solutions at the end of each booklet.

A good weekly routine, about two hours outside class:

1. Read the notes section for the session **before** it, quickly, to see the words.
2. Attend, and do the “try this now” exercise in class with a pen.
3. After the session, redo the worked examples in the notes without looking, then do the matching self-assessment questions.
4. Check one or two answers in NumPy.

PITFALL · DO THE CALCULATION YOURSELF

Every calculation in this course looks obvious when someone else does it on the board. The only test is whether you can do it on a blank page. Redo the worked examples yourself; that is where the learning happens.

Tools you will need

- A pen, and a notebook kept only for this course.
- **Git** and the **GitHub CLI** on your laptop, to submit through Classroom 50.
- **Python 3 with NumPy**, on your laptop, or **Google Colab** in a browser. The Getting Started guide covers Windows, macOS and Linux, and Colab.

NumPy is used for one purpose in this course: to **check** what you have calculated by hand. The examinations are on paper, with no computer.

Policies

Late submissions. Marked out of 80 percent within 48 hours; not accepted after that except with documented cause.

Collaboration. Discussing how to approach a problem is encouraged. Copying an answer, or submitting code you did not write, is not.

AI tools. Permitted for learning: asking for an explanation, a second worked example, or a hint. You must be able to explain every step of every answer and every line of code you submit. The viva is where that is checked, and a submission you cannot explain earns no marks for the part you cannot

explain.

Handwriting. Parts A and B are handwritten and scanned into one PDF, pages in order, readable. Illegible working earns no method marks.

Reading

The notes booklets are self-contained. If you want another voice on the same material:

- Martin Laforest, *The Mathematics of Quantum Mechanics*, Institute for Quantum Computing, University of Waterloo. The main reference; this course follows its three chapters.
- C. Hughes, J. Isaacson, A. Perry, R. F. Sun and J. Turner, *Quantum Computing for the Quantum Curious*, Springer, open access. Friendly, with more physical motivation.
- Y. Billig, *Quantum Computing for High School Students*. Useful for the qubit and measurement chapters.

Getting help

Bring the problem, what you did, where you think it went wrong, and what you have already checked. For a NumPy problem, bring the exact code and the exact output. A photograph of a page of working is fine; a description of it is not.